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Mahidol University

Biomedical Engineering

Undergraduate Student Handbook

Academic Year 2026–2027

Faculty of Engineering

Mahidol University August 3, 2026

Contents

Chapter 1

Introduction

1.1 Mahidol University

Mahidol University has been living a long history since the establishment of "Siriraj Hospital" in 1888, then developed to a medical school in the next year. The name was bestowed upon by His Late Majesty King Bhumibol Adulyadej after His Royal Highness Prince Mahidol of Songkla.

Mahidol University had continued to progress in all aspects to stay relevant and conform to the rapidly changing world and society. The significant role of the university is to produce quality graduates, who are knowledgeable in their chosen fields and mindful of morality, to serve as human capital for the country's current development. Moreover, the university has vouched for academic development, research support for innovations, improvements of university's physical systems and environment as an eco-university as well as internationalization. All strategies are aligned with Mahidol University's aspiration towards a world class university. The university's current milestone is being ranked as the top university in the region and No.1 in Thailand.

Mahidol University is recognized as a large higher education institution comprising of academicians and professionals in every field, both in arts and sciences. Therefore, the current university administration has set our target to maintain our status as the leading university in the country and the region. We are aspired to be the source of knowledge for the benefits of society and country as Mahidol University's slogan **"Wisdom of the Land"**.

Mahidol University Focus:

Innovative Education

We aim for our students to develop the skills they need to overcome global challenges, explore new opportunities, and grow as globally competent citizens. We focus on student centered and active learning so that all graduates have the professional skills they need for their work, as well as the creativity and adaptability necessary for the changing world.

Translational and Transformative Research

Our research supports innovation in technology, clinical practice, and social justice, and places our students in a cutting-edge research environment. Our research strategyfocuses on 3 key areas; Medicine and Health Sciences, Innovative Technologies,and Resilient Societies.This high level structure supports interdisciplinarycollaboration, and ensures that our findings contribute to social and economicdevelopment, locally and internationally.

Global Connectivity At Mahidol, we have over 450 partnership university and research institution around the world, allowing us to collaborate on a range of academic and research activities. Our aim for the future is to move beyond these agreements, to build strong relationships with selected Strategic Partners.

An Environment for Living and Learning Mahidol University has 3 campuses in and around Bangkok, as well as 3 provincial campuses to spread high quality research and education across Thailand. Our main Salaya campus, on the outskirts of Bangkok, has wide range of educational, scientific, sports, and social facilities across 520 acres with gardens, lakes, and lawns.

For the Benefit of Mankind Our work in education, research, and services is centered around what we have to contribute to society. Healthcare Services, Global Problems, Local Solutions, and Knowledge Sharing.

1.2 Mahidol Core Value

Mahidol University emphasizes a set of core values known as "MU Core Values." These values reflect the university's mission and vision and guide the behavior and decision-making of its community members. The Mahidol Core Values are:

Mastery: Commitment to excellence and continuous improvement in academic and professional pursuits.

Altruism: Dedication to the welfare of others and the betterment of society.

Harmony: Fostering unity, collaboration, and mutual respect among diverse groups.

Integrity: Upholding ethical standards, honesty, and transparency in all actions.

Determination: Demonstrating resilience, perseverance, and the drive to overcome challenges.

Originality: Encouraging innovation, creativity, and the generation of new knowledge and ideas.

Leadership: Cultivating the ability to inspire, influence, and guide others towards shared goals.

These core values form the acronym "MAHIDOL," representing the university's dedication to creating a positive impact on society through education, research, and community service.

Table 1.1: Mahidol Core Value

Core Value: Mahidol University		
M	Mastery	1. Self Directed with Self Awareness 2. Master technical competency in each professional specialty 3. Personal Learning 4. Agility: Think & Understand quickl
A	Altruism	1. Organization First 2. Customer-Focused Driven 3. Societal Responsibility
H	Harmony	1. Value Workforce Member 2. Empathy & Unity 3. Synergy
I	Integrity	1. Truthfulness 2. Moral & Ethics: Code of Conduct 3. Management by Fact
D	Determination	1. Commitment & Faith 2. Perseverance 3. Achievement Oriented & Creating Value
O	Originality	1. Courageous to be the Best 2. Driving for Future 3. Novelty & Innovation
L	Leadership	1. Calm & Certain 2. Influencing People 3. Visioning

1.3 Welcome from Biomedical Engineering Department Chair

Figure 1.1: Head of Department of Biomedical Engineering, Assoc. Prof. Warakorn Charoensuk

It is my distinct pleasure to welcome you to the Department of Biomedical Engineering. As the Chair of this esteemed department, I am thrilled to see so many bright and eager faces ready to embark on this exciting journey with us.

Biomedical engineering is a field that stands at the intersection of technology and healthcare, driven by the desire to improve human lives. Here in our department, we are not just a group of students and faculty; we are a family, united by a shared passion for innovation, discovery, and making a positive impact on the world.

One of the cornerstones of our program is the vast array of research opportunities available to you. Our department is home to cutting-edge laboratories and research centers, where you will have the chance to work alongside leading experts in the field. Whether your interest lies in tissue engineering, medical imaging, biomaterials, or any of the many other exciting areas of biomedical engineering, you will find ample opportunities to engage in groundbreaking research. We encourage you to dive in, ask questions, and contribute your unique perspectives and ideas. Research is not just about learning; it's about creating new knowledge and pushing the boundaries of what is possible.

As members of the biomedical engineering family, we support one another in our academic and professional endeavors. Collaboration and teamwork are essential elements of our culture, and we believe that by working together, we can achieve more than we ever could alone. You will find that your peers and professors are not just colleagues but also mentors and friends who are here to help you succeed. Take advantage of this supportive environment, build lasting relationships, and learn from one another.

In addition to fostering a strong community within our department, we are committed to preparing you to become global citizens. Biomedical engineering is a discipline that transcends borders, and the solutions we develop here have the potential to impact lives around the world. We encourage you to think globally, to understand the diverse needs and challenges faced by different populations, and to approach your work with a sense of responsibility and compassion. Whether through international collaborations, study abroad programs, or working on projects that address global health issues, you will have opportunities to broaden your horizons and make a difference on a global scale.

In closing, I want to remind you that you are part of something truly special here. The work you do will not only shape your future but also contribute to the betterment of society. Embrace the challenges, seize the opportunities, and know that you have the full support of your biomedical engineering family behind you.

Welcome to our department, and I look forward to seeing all that you will achieve.

1.4 Faculty of Engineering Mission Statement and Strategic Aims

The mission of Mahidol University's Faculty of Engineering is to lead in advanced engineering and technology through world-class research, innovation, and academics, contributing to the global society. Our vision focuses on interdisciplinary research and internationalized education towards achieving world-class engineering standards. We aim to have all graduate programs certified by the AUN-QA standard, ensure undergraduate programs are accredited by ABET, and continuously enhance the quality of Mahidol Engineering to be recognized globally. Along with University missions, Faculty of Engineering has a strategic plan in: i) Integrated Research into World-Class Engineering,; ii) International Education towards World-Class Engineering; iii) Academic Services for Sustainability; iv) Excellent Organizational Management Culture for Sustainability; v) Global Competitiveness

Chapter 2

Introduction to the Department

2.1 Welcome from Department

Welcome from Assoc.Prof. Dr.Benchaporn Lertanantawong,
Director of Undergraduate Study

Welcome to the Undergraduate Program in Biomedical Engineering! As the Director of Undergraduate Studies, I am thrilled to see so many bright, eager faces ready to embark on this exciting academic journey with us.

Our program is designed not just to educate but to inspire. Here, you will delve deep into the fascinating world where engineering meets the life sciences. You will learn the fundamental principles that underpin both fields and explore the innovative ways they intersect to solve real-world problems. From biomechanics and biomaterials to medical imaging and tissue engineering, your coursework will cover a broad spectrum of topics, providing you with a solid foundation in biomedical engineering.

But learning doesn't stop in the classroom. We believe that exploration is key to a well-rounded education. You will have countless opportunities to engage in research projects, internships, and hands-on activities that will allow you to apply what you've learned and gain practical experience. Our state-of-the-art laboratories and research centers are at your disposal, and we encourage you to dive in, ask questions, and push the boundaries of your knowledge.

Our program also emphasizes the importance of extracurricular activities. Joining student organizations, participating in competitions, and attending workshops and seminars will enrich your academic experience and help you develop essential skills such as leadership, teamwork, and communication. These activities will also allow you to build a network of peers and mentors who will support you throughout your academic journey and beyond.

So, what can you expect from our undergraduate program? You can expect to be challenged and inspired. You can expect to learn from some of the brightest minds in the field. You can expect to grow both academically and personally. And most importantly, you can expect to be part of a vibrant, supportive community that is committed to helping you succeed.

As you look to the future, know that a degree in biomedical engineering opens doors to a wide range of exciting career opportunities. Whether you choose to pursue further studies, work in industry, or engage in research, the skills and knowledge you acquire here will be invaluable. Biomedical engineers are in high demand, and the work you do has the potential to make a profound impact on society.

In closing, I encourage you to take full advantage of everything our program has to offer. Be curious, be proactive, and never hesitate to seek out new experiences. Your time here will be what you make of it, and I have no doubt that you will achieve great things. Welcome to the Biomedical Engineering family. I look forward to witnessing your journey and celebrating your successes.

Detailed arranagements are subject to frequently changes, often introduced by university registry and faculty of engineering. For up-to-date information please consult the department administration team or web-pages:

https://www.eg.mahidol.ac.th/dept/egbe/weben/

We are sure that you will enjoy your time here and will contribute substantially to developing the subject.

Department of Biomedical Engineering is part of the Faculty of Engineering, which encompasses all the different types of Engineering department withing the university. Relevant information about the Faculty of Engineering can be found here.

- General information about Faculty of Engineering
  `https://www.eg.mahidol.ac.th/egmu_eng/`
- Research activities support and facilities within the faculty
  `https://www.eg.mahidol.ac.th/egmu_eng/research-and-services/research/research-fields`
- Faculty team members
  `https://www.eg.mahidol.ac.th/egmu_eng/faculty/faculty-by-name`

2.2 Department Structure, Academic and Administrative Staff

The Undergraduate study team of the Department of Biomedical Engineering is composed of various academics and administrative staff. These include Director of Undergraduate study, Undergraduate tutor, one for each year group, Academic Progress Tutor, and Program administrator. The current Undergraduate team's members are specified in table 2.1 and 2.2.

additional information about academic and adminstrative staff of the Department of Biomedical engineering can be founded on Department Website:

https://www.eg.mahidol.ac.th/dept/egbe/weben/faculty.html

Table 2.1: Undergraduate Study Team members

Role	Name	Photo
Director of Undergraduate Study	Assoc. Prof. Benchaporn Lertanantawong	
Academic Tutor First Year BME#20	Dr. Shen Treratanakulchai	
Academic Tutor Second Year BME#20	Dr. Chalaisorn Thanapongpibul	
Academic Tutor Third Year BME#20	Dr. Yuttana Itsarachaiyot	
Academic Tutor Fourth Year BME#20	Dr. Titipat Achakulvisut	
Academic Progress Tutor	Dr. Pornpat Athamanolap	
Student Activities Mentor	Dr. Pinunta Nittayacharn	
Program Committee	Assoc. Prof. Panrasee Ritthipravat	
	Dr. Pracha Yambangyang	
	Dr. Jetsada Arnin	

2.3 You and your mentor

The primary responsibility for organizing your study lies with you. However, it is your academic tutor's responsibility to guide your study, point you in interesting directions, monitor your progress and generally provide moral and technical support. Tutor differ in their methods but you should normally expected to see your tutor at least once fortnight. Feel free to contract him or her at any time if you have a problem or are unsure how to proceed.

you will find that you can obtain the most benefit from meetigns with your tutor if you prepare some material for him or her to read or formulate some specific questions you would like to discuss. Your academic progress tutor is to give help when your tutor is unavailable and generally to keep in tour with your progress.

Undergraduate student should do your best to participate as fully as possible in the academic life of the Department. You will find that informal discussions with other students or researchers play a large part in your education. You should work hard to build up your relationship with your project supervisor, but it dows occasionally happen that you find it impossible to work together. Or your research and project is not aligned with project supervisor or out of interested area. In such a circumstance a change of your project supervisor is appropriate and can sometimes be arranged. Discuss this matter with your tutor if you can, and then involve further with the director of programme.

Undergraduate Academic tutor The director is responsible for the overall running of the undergraduate study and for providing pastoral support for student. She also available to discuss any matter, personal or academic, in confidence. Additionally, the Academic progress tutor is another person who can provide counsel, encouragement, career advice, and generally take an interest in your development. The academic progress tutor is independent of your tutor and takes no responsibility of performance and assessment of your progress. Sometime, he or she will organise social event for their year group.

2.4 Adminstrative Staff

The Undergraduate administrative staff will help and support your study. They manage and organize student records, ensuring that academic achievements and personal information are accurately maintained and easily accessible. They also coordinate scheduling and registration, making it easier for students to enroll in classes and access necessary resources. Additionally, they facilitate communication between students and faculty, helping to address any concerns or issues that arise. By overseeing the logistical and operational aspects of the educational environment, administrative staff create a structured and supportive framework that allows students to focus on their studies and personal growth. Below is the table 2.2 of Administrative staff in our department.

2.5 Department Information

General information about the department can be found on the department website https://www.eg.mahidol.ac.th/dept/egbe/weben/overview.html

Following the various links you will be able to access to information about

Table 2.2: Undergraduate Study Administrative Team members

Role	**Name**	**Photo**
Purchasing and Human Resource	Vipada Chuairaksa (Nong)	
Undergradaute Study Adminstrator	Wachiraporn Thongkum (Mutt)	
Graduated Study Adminstrator International Relations	Matchima Rattanalam (Matshi)	
Public Relations	Inchalita Suwanrangsima (Beam)	
Mechanical and Maintainance Engineer	Saichol Buachan (Chao)	
Computer and IT Specialist	Narluchar Wimolluck (Ake)	
Electrical Engineer	Puttipong Sonkiew (Nun)	

- Research activity in the department
  https://www.eg.mahidol.ac.th/dept/egbe/weben/research_track.html
- Public events and Research Seminar organised and run by member of staff of the department
  https://www.facebook.com/bmemahidol/
- News and Events within Faculty of Engineering
  https://www.eg.mahidol.ac.th/egmu_eng/about/news-events/previous-news

Research seminars and social activities

Research is a social activity. You will learn more by talking with people working in your area and attending seminars than you will by reading papers in isolation. The department has a wide range of active research group that are well. We are also encourage the student to participate a research competition, project pitching, and other engineering project to give you experiences.

There frequently informal seminars to complement the official Departmental seminars (held usually on Tuesday Morning). Seminars are announced by email, facebook channel, and on the Department's webpages. As a biomedcial engineering student you are entitled to attend seminars and use any departmental facilities such as the common room on the 1st Bay arena of the first building, Innogineer studio as a make studio to create your engineering prototype, and other lab area to discuss and build up your engineering project. Try and maximise the opportunities this offers. The undergraduate adminstrator will reach out and notice the event to all the student in the Department.

2.6 Laboratory Code of Conduct

Welcome to the department lab! To ensure a safe, productive, and collaborative environment, please adhere to the following guidelines:

1. **Safety First**
    - **Follow Safety Protocols:** Familiarize yourself with and adhere to all lab safety protocols. Always wear appropriate personal protective equipment (PPE), such as lab coats, gloves, and safety goggles.
    - **Know Emergency Procedures:** Learn the locations of emergency exits, fire extinguishers, eye wash stations, and first aid kits. Be aware of the emergency contact numbers. For medical or general emergencies at Mahidol University (Salaya Campus), call 1669 for national medical emergency services or the campus security center at 02-441-4400 (press 0) or 02-441-9318.
    - **No Food or Drink:** Consumption of food or beverages in the lab is strictly prohibited to prevent contamination and accidents.
    - **Dress Code: Lab Coats** - Always wear a lab coat to protect your clothing and skin from spills and splashes. **Long Pants and Closed-Toe Shoes**- Wear long pants and closed-toe shoes to protect your legs and feet. Sandals, shorts, and skirts are not permitted. **Tie Back Long Hair**- If you have long hair, tie it back to prevent it from coming into contact with chemicals, flames, or equipment. **Avoid Loose Clothing and Jewelry** - Loose clothing

and dangling jewelry can get caught in equipment or come into contact with hazardous materials. Avoid wearing them in the lab.

2. **Follow Safety Protocols:** Familiarize yourself with and adhere to all lab safety protocols. Always wear appropriate personal protective equipment (PPE), such as lab coats, gloves, and safety goggles. Know Emergency Procedures: Learn the locations of emergency exits, fire extinguishers, eye wash stations, and first aid kits. Be aware of the emergency contact numbers.

3. **Lab Conduct:**

   - Be Respectful-Treat all lab members with respect and maintain a professional attitude. Collaboration and communication are key to a successful lab environment.
   - Stay Organized - Keep your workspace tidy. Clean up after yourself and properly dispose of any waste materials. put all the equipment where it belong after used.
   - Label Everything: Clearly label all samples, reagents, and equipment with your name, date, and any other relevant information.

4. **Equipment and Materials**

   - Ask for permission with the researcher in the lab before using anything. You are not allowed to train to use the machine or equipment other user to other student unless you are responsible for the training.
   - You are not allowed to use the machine for other student without permission.
   - Use Equipment Properly: Only use equipment that you have been trained to operate. Report any malfunctions or issues to the lab supervisor immediately.
   - Conserve Resources: Be mindful of the materials and resources you use. Avoid unnecessary waste and share supplies whenever possible. Proper Storage: Store chemicals, reagents, and samples in their designated areas. Ensure all containers are properly sealed and labeled.

5. **Data management**: Keep detailed and accurate records of all experiments, observations, and results in your lab notebook. Digital data should be regularly backed up. Respect the confidentiality of all lab data and intellectual property. Do not share sensitive information without proper authorization.

6. **Communication**: Ask Questions: If you are unsure about any procedure, equipment, or protocol, ask a more experienced lab member or the supervisor for assistance. Regular Meetings: Attend all scheduled lab meetings and actively participate in discussions. This is a great opportunity to share your progress and learn from others.

7. **Compliance**: Follow Protocols: Adhere to all lab protocols, standard operating procedures (SOPs), and institutional guidelines. Ethical Conduct: Conduct all experiments and research with integrity and honesty. Any form of data manipulation or misconduct will not be tolerated.

Remember that this is the standard protocol for the lab usage and safety. We run this lab conduct to ensure your safety at first. And please remember that each lab has also their style and their lab code of conduct, please follow their guideline strictly. **if you don't follow the lab guideline, you could be revoke for lab access in the future.** By following these guidelines, you help maintain a safe, efficient, and harmonious lab environment. Welcome aboard and happy experimenting!

2.7 Research Opportunities

Participating in lab research offers students an invaluable opportunity to immerse themselves in the scientific process and gain hands-on experience that goes beyond classroom learning. By engaging in lab work, students can deepen their understanding of scientific theories and methodologies, apply their knowledge to real-world problems, and develop critical thinking and problem-solving skills. This practical experience is essential for mastering advanced techniques and understanding the intricacies of experimental design and data analysis. Moreover, lab research fosters collaboration with experienced researchers and peers, providing a platform to learn from others and contribute to innovative projects. These experiences not only enrich academic knowledge but also open doors to future career opportunities in diverse fields such as academia, industry, biotechnology, and healthcare.

Attending the lab regularly demonstrates a student's commitment and initiative, qualities that are highly valued by potential employers and graduate programs. It helps build a professional network, improves technical skills, and provides exposure to cutting-edge research and technologies. Additionally, lab research can lead to opportunities for presenting findings at conferences, co-authoring publications, and participating in grant-funded projects, all of which are impressive additions to a resume or graduate school application. Embrace this chance to grow both personally and professionally, contribute to meaningful scientific advancements, and pave the way for a successful future career in science and technology. You can have a look on the list of the lab in the chapter 4.

2.8 Student's Scholarship and other fund opportunities

Mahidol University has policy to support scholarship for students who lack of financial support but outstanding in learning with good behavior and other supports. The source of scholarship consists of

- **Internal funds (Please contact division of student affairs from your college or your faculty)**:

    1. King Phumibol's scholarships for students who are outstanding in learning and activities but lack of financial resources. This scholarship worth 50,000 THB and give on the graduation ceremony.
    2. Mahidol University's scholarships for students who have good behavior but lack of financial resources. This scholarship worth 5,000-10,000 THB.
    3. Scholarships from benefits of donors for students who outstanding in learning but lack of financial support.

- **Government and private company funds:** are external funds sources from government office / bank / private company / shop / foundation / charity and donors. University will use this scholarship as the objectives of donors such as the characteristic of scholarship receiver.
- **University emergency funds**: for students who had a serious accident that effected their studies such as their parent pass a way or their house burned down. Please contact Division of Student affairs from your college or faculty
- **Loan funds for students in needed**: such as accidently lost their money, their parent cannot afford money in time and they need to use the money as soon as possible. Please contact Division of Student affairs from your college or faculty. Students can loan between 500-5,000 THB and must pay back within 6 months.
- **Loan funds for education**: The objective of Loan funds for education is to provide loans for students who lack of financial supports to pay for their tuition fees, everything relate to education and living expenses. Loan funds consist of 2 types as follows:
  1. SLF type https://op.mahidol.ac.th/sa/studentloan for university tuition fees but not exceed the funds limitations and living expenses total 26,400 THB per year.
  2. ICL type https://op.mahidol.ac.th/sa/studentloan) for university tuition fees but not exceed the funds limitations but loaner must study in main demand programs and clarity of work force production according to the fund committee's announcement.

More details about these awards are available on the Mahidol University scholarship website. Additional support includes undergraduate mobility funding, described on the undergraduate scholarship and mobility webpage, and funding for conference participation.

Note that all the scholarship or funding has their eligibilty criteria, please learn more about the scholarship. The department will announce any scholarship opportunities through your email.

(a) Front of Biomedical Engineering Common Room

(b) Biomedical Engineering Common Room Area

Figure 2.1: Department Common Room and Workshop Area

2.9 Additional Department Information

- **Common Room**
  At the faculty of engineer, there are a common space for Engineering student which are at the ground floor of the Engineering Building 1. For the biomedical engineering student, there is the new area of common room in the Bay Arena 1, Building 3. In this is area you can have an access to other department laboratory or other workshop. The student in encourage to use this common room. This building is normally open from 09:00 to 17:00(New time table will be announced later).

- **Co-Working Space for Student- Innogineer Studio**
  Not just the common room area that you are welcome. There is other area in the faculty that you can stay and work with your friends. Innogineer Studio Center aims to create "Innovative Society and Entrepreneurship" allows Thailand to increase product value through innovation, from SMEs to startups, cultivating knowledge and cultivating Entrepreneurial Mindset and technical and hand-on skills to students, makers, start-up engineers and researchers in various areas of freedom of thought, imagination in the creation of a joint center Innogineer Studio, with a total area of over 800 square meters and advanced technology consists

  1. Mechanical Studio complete with Milling Machine , CNC Machine

Figure 2.2: Innogineer studio

2. Electric Studio advanced with equipment such as Electronic Supplier, Microcontroller, Oscilloscope, Power Supply, Function Generator
3. Assembly Studio, such as 3 D Scanner high resolution with a pin head with a laser scanner (3 D Laser Scanning Arm the System the CMM)
4. Prototyping Studio, such as 3 the D a Printer, and molded plastic 3 the D Machine Studio, consists of a metal cutting lathe
5. Gallery Room displays space for invention and innovation
6. Co-Working space, the Area for learning and innovation, accommodating 30-40 people.
7. Meeting Room, A meeting room area that can accommodate 20-30 people, with audio equipment and LCD projectors available, including Innogineer Studio Shop for showcasing various innovations.

You can subscribe this website for further detail Innogineer Studio Facebook or contract email to mueg.innogineerstudio@gmail.com or call Tel. +662-889-2138 ext 6163

Mail and contract information
Students must provide their contract information upon registration at the start of the each academic session. It is essential that any subsequent changes of address are notified to both Registry and the Department Programme Administrator immediately.

Telephone and Fax
Student can dial the department phone by using this number. This number is a direct number to Department administration team.
Tel: +66-2-441-4255 or +662-889-2138 Ext 6351-2.
Fax: +66-2-441-4254 or +662-889-2138 Ext 6366

Identify Cards
Your student ID card is an smart card that hold your study information. It is a smart card that link you wiht the Bank account, SCB Bank. So, please keep

in mind that it is your own responsibility to keep it safely. In case of new students who have never had a Siam Commercial Public Company Limited account or want to open a new bank account. You can download the 0 Baht account opening form (Zero Baht). To apply for a SMART ID CARD (debit card), please visit https://smartedu.mahidol.ac.th by logging in to the system and entering the Student Information menu. Select to download the 0 Baht account opening form (Zero Baht Total) and contact the nearest branch of the Bank. If it is not convenient, you can open an online account at "SCB EASY APP". Please see the full detail in this website https://mustudent.mahidol.ac.th/newstudentidcard/

Out of hours access
Normal university hours are between 08:00 to 17:00, Monday to Friday. The times outside these hours are known as 'out of hours' period. Student are permitted to work (Not play or sleep) in some part of the building. Entry of the building and laboratory area will be not permitted after 22.00 and all students must leave the building by 23.00. However, if you have an important lab work to conduct, you have to ask the permission of the lab leader and has someone with you through out. All the student must carry ID Card to be allowed in the area.

Emergency Calls For medical or general emergencies at Mahidol University (Salaya Campus), call 1669 for national medical emergency services or the campus security center at 02-441-4400 (press 0) or 02-441-9318. For 24-hour student mental health or urgent crisis support, call the MU Hotline at 088-874-7385.

2.10 Immigration VISA for overseas student

International students require a Non-Immigrant Education Visa (Category "ED") to study in Thailand. Whenever possible, applicants should apply for their student visa well in advance of departing for Thailand. Applications should be made in person at the Royal Thai Embassy or Consulate-General in their home country. Internation student must follow the guideline from Royal Thai Embassy or Consulate-General in their home country. There are 3 typical requirement you need to check:

1. **90-day Reporting**
   International students must report to their local immigration office every 90-day (up to 7 days before or after the 90-day period expires). Failure to report within the specified timeframe will result in a fine of 2000 Thai baht (US$ 67). Students can report at their local immigration office in person by proxy, or via registered mail. If you would like to notify immigration via registered mail, you will need to send the following documents to your local immigration office, making sure they arrive 7 days before the 90 day period expires.

2. **Visa extension**
   International students must extend their visa no later than 3 months after their arrival. The visa should also be extended at least 15 days in advance of its expiration date. Ignoring the extension deadline will result in fines of 500 Thai baht (US$ 17) per day [up to a maximum of 20,000 Thai baht (US$ 667)]. International students can contact the International Relations Division to request an official letter for their

visa extension, by submitting a copy of their passport. Following this, they should submit the following documents in person at their local Immigration Office:

3. **Re-entry Permit**
   If you have a single entry visa and would like to travel outside of Thailand for a short period during your studies, then you will need to apply for a single or multiple re-entry permits before departure. Please contact the Immigration Bureau to apply for a Re-entry Permit.

Understanding and complying with immigration and visa requirements is essential for a smooth and successful international study experience. It's advisable to stay informed about any changes in immigration policies and seek guidance from the international student office at your institution or a qualified immigration advisor. for full detail please visit https://mahidol.ac.th/visa/ or contract matchima.rat@mahidol.edu

Departmental Induction
during the first week of your bachelor degree study various events are run within the department.
These include:

- welcome meeting from faculty of engineering run by Undergraduate Study team who will introduce you and give you the relative information on the main administrative tasks that you are required to do at the start of your study, such as safety induction section.
- Welcome meeting with BME department run by Academic team, who will introduce you the relevance aspect of your study course at Mahidol University, useful links and milestone schedule. This talk is also attended by lecturer in department, Undergraduate tutor who you will have the pleasure to meet and talk.

information of the day, time and location of these events is provided to you as part of your welcome pack.

Chapter 3

Curriculum

3.1 General Curriculum Information

The undergraduate study has a program title as
"Bachelor of Engineering Program in Biomedical Engineering"
The degree offered and Field of Study is *Bachelor of Engineering (Biomedical Engineering), B.Eng.*

There are 3 difference study plan for your choice:

- **Plan A: Regular Program**
  The number of credits required for the program is at least 131 credits.
- **Plan B: Advanced Academic Program**
  The number of credits required for the program is at least 141 credits.
- **Plan C: Dual Degree Program**
  The numbers of credits from Mahidol are not less than 78 credits and the numbers of credits from University of Strathclyde are not less than 240 credits.

The study program has different requirement in each offered program. You will have to register for study course following the plan as shown in table 3.1. This Curriculum complies with the Standard of Undergraduate Programs of Study announced by the Ministry of Education 2024. Courses included the course from various faculty and department. For the full course in each year, you can have a look at the table 3.2. You can recognize by its course code. For example EGXX, The first two letter describe that this course is subjected to faculty of engineering credit. the last two letters described subject categories. For the general education, It is not necessary that it need to be only GEXX course.
Note that: Students must register the General Education courses not less 1 credit for each Literacy group consisting of (1) MU Literacy, (2) Health Literacy, (3) Science and Environment Literacy, (4) Intercultural and Global Awareness Literacy, (5) Civic Literacy, (6) Finance and Management Literacy. Students have the choice of completing the General Education courses provided by other programs/departments/faculties. By doing so, this is to fulfill the credit requirement under the consent of advisor, the Program Director or Curriculum Committee in accordance with Mahidol University's regulations. For the full detail of the offering course, please check this website https://clil.mahidol.ac.th/

For the Advanced Academic Programming, you have to register more courses in third or fourth year to fulfill program requirement. The courses that you have to register are graduated courses in our master degree and philosophy degree in Biomedical Engineering. This program required you to conduct the fourth year capstone project as standalone project. It is means that once you enter this program, you are eligible to continue study in our master degree program and finished the course work within first year. Consequently, you can continue and develop further your fourth year project as a master degree thesis within one year to complete.

For the Dual Degree Program, Students take courses at Mahidol University in their first and second years of the program. Once they complete their second year with at least 78 credits followed the detail show in the table 3.1, they are eligible to continue their third and fourth years at University of Strathclyde. Students must complete at least 240 Strathclyde credits.

Note that: please seek advice with your tutor and Undergraduate study administrator.

Table 3.1: Curriculum for Undergraduate study in Biomedical Engineering

Courses	**Credits**		
	Regular Program (Plan A)	**Advanced Academic Program (Plan B)**	**Double Degree Program (Plan C)**
1) General Education no less than (MU Literacy, Health Literacy, Science and Environment Literacy, Intercultural and Global Awareness Literacy, Civic Literacy and Finance and Management Literacy)	24	24	14 (MU)
2) Core courses no less than	**101**	**111**	**64 (MU) + 180 (UoS)**
- Core Courses	30	30	30 (MU)
- Required Courses	62	62	34 (MU) + 180 (UoS)
- Elective Courses	6	18	-
- Engineering Training	3	3	-
3) Free Electives	**6**	**6**	**60 (UoS)**
Total credits	**131**	**141**	**78 (MU) + 240 (UoS)**

At the end of third year, student will be notified that what research group they have been selected. Therefore, student will be allocated in different study plan due to their research. On the fourth year, students will select at least 6 credits from the following category of courses according to the track of their senior project. Students can also select courses outside the track with the permission from the department. The department will update the list of courses to students. Here are three tracks and recommendation for the study based on their research project. Finally, there might be an alternative course that could you could take for the full course detail please seek advice with your undergraduate administrator.

Table 3.2: Curriculum for Undergraduate Study

Year 1					
1st Semester			**2nd Semester**		
Course Code	**Credit**	**Course Name**	**Course Code**	**Credit**	**Course Name**
SCMA 101	2	Mathematics 1	SCMA 102	4	Mathematics 2
SCBE 102	1	General Biology Lab 1	SCPY 162	3	General Physics 2
SCPY 111	1	Physics Lab.	SCCH 172	3	Organic Chemistry
SCPY 161	3	General Physics	EGBI 101	2	Basic Engineering Skills in BME
SCCH 161	3	General Chemistry	EGBI 110	3	Engineering Materials
SCCH 169	1	Chemistry Lab.	EGBI 120	3	Engineering Drawing and Computer Aided Design
SCSL 190	3	Wonderful Life		2	General Education Elective
EGBI 100	1	Biomedical Engineering in the Real World			
EGBI 122	3	Computer Programming			
	2	General Education Elective			
Total	**20**		**Total**	**20**	
Year 2					
1st Semester			**2nd Semester**		
Course Code	**Credit**	**Course Name**	**Course Code**	**Credit**	**Course Name**
SCAN 201	3	Essential Human Anatomy	EGBI 201	1	Biomedical Engineering Lab 1
EGBI 202	3	Engineering Mathematics	EGBI 220	3	Computational Methods in BME
			EGBI 221	3	Biostatistics and Probability
EGBI 234	3	Electric Circuits for Biomedical Engineering	EGBI 223	3	Digital Systems and Microprocessor
EGBI 252	3	Biochemistry and Molecular Biology	EGBI 233	3	Electrical and Electronics in Medicine
EGBI 260	3	Biomechanics 1	EGBI 261	3	Biomechanics 2
EGBI 270	3	Biomedical Thermodynamics		2	General Education Elective
	2	General Education Elective			
Total	**20**		**Total**	**18**	
Year 3					
1st Semester			**2nd Semester**		
Course Code	**Credit**	**Course Name**	**Course Code**	**Credit**	**Course Name**
EGBI 300	1	Biomedical Engineering Lab 2	EGBI 331	3	Control System for Biomedical Engineering
EGBI 301	3	Design for Biomedical Engineering	EGBI 351	3	Biocompatability
EGBI 330	3	Biomedical Measurement and	EGBI 390	3	Business for Medical Entrepreneur
EGBI 340	3	Biomedical Signals and Systems		3	Free Elective(EGBI 401: Medical Imaging)
EGBI 352	3	Biomaterials		2	General Education Elective
	3	Free Elective(EGBI 405: AI in Medical Applications)		2	General Education Elective
			Summer		
			EGBI 391	3	Biomedical Engineering Training
Total	**16**		**Total**	**19**	
Year 4					
1st Semester			**2nd Semester**		
Course Code	**Credit**	**Course Name**	**Course Code**	**Credit**	**Course Name**
EGBI 495	3	Biomedical Engineering Capstone Project 1	EGBI 496	3	Biomedical Engineering Capstone Project 2
EGBI 4XX	3	Elective Courses	EGBI 4XX	3	Elective Courses
GE XX	3	General Education Elective	GE XX	3	General Education Elective
Total	**9**		**Total**	**9**	

- **Neuroengineering and Medical Imaging**
  1st Semester
  - EGBI 441 Medical Signal Processing

  2nd Semester
  - EGBI 444 Neuroengineering
- **Biomaterials and Biosensors**
  1st Semester
  - EGBI 483 Drug Delivery System

  2nd Semester
  - EGBI 485 Biosensors
- **Medical Robotics and Instrumentations**
  1st Semester
  - EGBI 481 Introduction to Medical Robotics

  2nd Semester
  - ~~EGBI 403 Artificial Organs~~ TBA new subject

Note that: One more thing to check every time you register: look at each course's prerequisite requirements before you sign up. Many subjects—especially in your engineering core—are arranged in sequence, meaning you must complete and pass the prerequisite course first before the system will let you register for the next one that builds on it. So don't assume you can take an advanced subject just because it fits your schedule; confirm that you've already passed everything it requires. Plan your semester in the right order, because failing or skipping a prerequisite can block you from registering for the courses that depend on it and delay your progress toward graduation. If you're unsure whether one course is a prerequisite for another, check the course description or ask your academic advisor before you finalize your registration.

3.2 Milestone and Department Requirement

The university and the Department impose certain formal milesone, which are outline here. The Department strongly advices student to plan for completion of the study by 4 years. If you are worried about your progress, talk to your supervisor, mentor or undergraduate tutor. Here is general outline:

timetable

Year 1

- Adjust to College Life: Learn how to manage time, study effectively, and seek help when needed. Build a Strong Foundation: Focus on mastering the basics in math, science, and introductory engineering courses. Develop Good Study Habits: Establish routines that include regular study times, staying organized, and utilizing campus resources like tutoring centers.

- Explore Interests: Join engineering-related clubs and organizations to explore different fields of engineering. Network: Start building a network with peers, professors, researcher in the lab, and advisors.
- At the end of your first year you should experience university and department activities. Try to explore all the research group in the department or even sit in and apprentice in some lab project. This will benefit to you for early research and engineering skills to fit your future career.

Year 2

- Major-Specific Courses: Start taking more courses related to your specific engineering discipline. Maintain GPA: Keep up with coursework and aim to maintain a strong GPA for future opportunities.
- Internships/Co-ops: Apply for summer internships or co-op programs to gain practical experience. Career Fairs: Attend career fairs and networking events to meet potential employers and learn about job opportunities. Skills Development: Take workshops or online courses to develop skills like programming, CAD, or other relevant tools.
- at the end of your second year you should have identified quire closely the area you wish to work in and have developed some ideas on which research field could be based.
- some of student who may interest in dual degree program should seek the advice from your tutor and adminstrator team to plan out your study.

Year 3

- This year is very important for your study. because you will have to submit an interest for research group selection. It would be benefit for leader in the research group and yourself, if you frequently attend the research lab. The selection process is not only based-on your grade and skill. It is also involve with the past research experience.
- At the end of year 3, you will have to register for engineering internship which you will experience your future career and training on specific skill in industry, research lab., start-up company, and national or internatioanl organization.
- Get involved in undergraduate research if interested in pursuing a deeper understanding of your field.
- Capstone Projects: Start planning for or participating in capstone projects or design competitions. Mentorship: Seek mentorship from faculty and industry professionals.
- Internships/Co-ops: Continue to seek internships or co-op opportunities. Networking: Strengthen your professional network through industry events and alumni connections. Certifications: Consider earning certifications relevant to your field.

Year 4

- Capstone Project: Focus on completing your capstone project, which integrates the knowledge and skills you've gained. Electives: Take advanced electives that align with your career goals

- Final Leadership Roles: Take on any final leadership roles in organizations to solidify your experience. Professional Conferences: Attend and, if possible, present at professional conferences.
- Job Search: Actively apply for full-time jobs or graduate school if that is your path. Career Services: Utilize career services for mock interviews, resume reviews, and job search strategies. FE Exam: If applicable, prepare for and take the Fundamentals of Engineering (FE) exam. Graduate School Applications: If considering further education, prepare and submit graduate school applications.
- at the beginning of your fourth year, you should prepare document to summary your study and build-up your fantastic resume. because sometime there is an opportunity call no surprise or even the study scholarship.
- If you able to publish scientific article during your study, it would a great push-up for your application in further study.

General Tips Throughout All Years

- Stay Curious: Continuously seek to learn and explore new technologies and advancements in your field.
- Build Relationships: Form strong relationships with professors, peers, and industry professionals.
- Balance: Maintain a balance between academic, professional, and personal life to ensure overall well-being.

Term-time Here is a typical term timetable

Table 3.3: Term Time Table

	1st Semester	**2nd Semester**	**Summer**
Open Semester	Week 1 Aug	Week 1 Jan	Week 1 June
Mid-Term Exam	Week 1 October	Week 1 March	-
Final Exam	Week 1 December	Week 1 May	Week 4 July
Close Semester	Week 2 December	Week 2 May	Week 4 July

3.3 Philosophy, Significance and Objectives of the Program

Philosophy, rationale, and objectives of the program To produce graduates based on research-based and outcome-based education. Graduates will have skills and knowledge in the field of Biomedical Engineering that are fundamental to career-long professional competence and meet current and emerging Biomedical Engineering needs. Integration and application of knowledge to benefit in accordance with the needs of society and the country with ethics in the professions and a responsibility on society.
Program Objectives:

- Apply advanced theoretical knowledge and practical skills to solve complex biomedical engineering problems.
- Design innovative biomedical solutions that address global healthcare needs, integrating AI and IT technologies.
- Communicate effectively and collaborate with diverse stakeholders in international contexts.
- Engage in lifelong learning to evaluate and adapt to emerging technologies and practices in biomedical engineering.
- Analyze and comply with international regulations and ethical considerations in biomedical engineering.
- Demonstrate leadership and entrepreneurial skills in advancing biomedical engineering initiatives.

Program–level Learning Outcomes: PLOs
On successful completion of this program, graduate will be able to:

PLO1 identify, formulate, and solve complex problems in the biomedical engineering area by applying principles of engineering, science, and mathematics.

PLO2 apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

PLO3communicate effectively with a range of audiences.

PLO4 recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

PLO5 function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

PLO6 develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

PLO7 acquire and apply new knowledge in biomedical engineering and/or related engineering disciplines using appropriate learning strategies.

PLO8* for Advanced academic program function independently and self-directed to explore knowledge and technology in biomedical engineering and/or related engineering discipline for advanced research.

3.4 Study Requirement for Graduation

Not only you have to complete course requirement from university standard and program requirement. There are several other requirements that you need to complete before the end of your study.

3.4.1 Course Completeness

Plan A: Regular program According to the regulations of study in diploma and bachelor degree of Mahidol University in 2024.

- students have to complete their credits as stated in the curriculum.
- Students must have a minimum 2.00 CUM-GPA
- Student must possess good behavior appropriate to the honor of bachelor of engineering.

Plan B: Advanced academic program According to the regulations of study in diploma and bachelor degree of Mahidol University in 2024.

- Students have to complete their credits as stated in the curriculum.
- Students must have a minimum 3.50 CUM-GPA
- Student must possess good behavior appropriate to the honor of bachelor of engineering.

Plan C: Dual degree with University of Strathclyde According to the regulations of study in diploma and bachelor degree of Mahidol University in 2024.

- Students have to complete their credits as stated in the curriculum.
- Students must have a minimum 2.00 CUM-GPA
- Student must possess good behavior appropriate to the honor of bachelor of engineering.
- Satisfy English language proficiency requirements according to the agreement between Mahidol University and University of Strathclyde.
- In the first and second year, students must complete the study at Mahidol University with credits per the agreement between the Faculty of Engineering, Mahidol University and the University of Strathclyde.
- In the third and fourth year, students must complete the study at the University of Strathclyde with credits per the agreement between the Faculty of Engineering, Mahidol University and the University of Strathclyde

  Note: The University of Strathclyde (hereinafter UoS) and Mahidol University (hereinafter Mahidol) agree to collaborate in arrangements whereby students from Mahidol can be admitted to degree courses at the UoS. Applicants, who have undertaken a relevant course of study at Mahidol, will be required to meet minimum entry standards as set out below in order to be admitted into UoS undergraduate degree courses.

  - Satisfactory completion of 2 years of study on the relevant BEng degree programme at Mahidol with a minimum of 70% average.
  - Mahidol applicants satisfy Strathclyde's English language proficiency requirements. These entry requirements are normally IELTS 6.0 (with no individual component below 5.5)

- Start date: September each year
- Duration at Strathclyde: Two years / 4 semesters

In the first and second year, students must complete the study at Mahidol University with credits per the agreement between the Faculty of Engineering, Mahidol University and the University of Strathclyde. In the third and fourth year, students must complete the study at the University of Strathclyde with credits per the agreement between the Faculty of Engineering, Mahidol University and the University of Strathclyde. Graduates will receive degree from University of Strathclyde after finish all of coursework's and other requirements of the programs.

3.4.2 Engineering Training - Internship

The third year students will register for Biomedical Engineering Training (EGBI 391) in the summer semester. We would like you to experience in real-life job for your future career. This is not the limit to any other field. You should discuss with your supervisor for the internship. We also would like you to distribute across the training area. At the end of the internship, you have to fulfill some requirement for training and also you need to present your internship at the beginning of the first semester at year 4. This is internship can take place domestically and internationally. *Below is the area of internship suggestion*

- Hospital and Clinical Training
  For example the hosipital for traning are King Chulalongkorn Memorial Hospita, Bhumibol Adulyadej Hospital, Bangkok Hospital, Rajavithi Hospital, Bumrungrad International Hospital, RamaHospital etc.

- Industrial training
  For example the company for traning are National Healthcare System Co., Ltd., Supreme Products Co. Ltd., Medtronic (Thailand) Limited, Johnson & Johnson (Thailand) Ltd., Olympus (Thailand) Co., Ltd., Filtech Enterprise 1994 Public Company Limited, BNH Medical Center Co., Ltd., B. Braun (Thailand) Co.,Ltd. etc.

- Research centre in government agency, private research lab
  For example the research center for training are Chulabhorn Research Institute, Prasat Neurological Institute, Food and Drug Administration, Queen Sirikit Medical Center, Singapore Institue for Neurotechnology (SINAPSE), National Science and Technology Development Agency etc.

- University research laboratory
  For example the university for traning are Harvard Medical School, National University of Singapore, Medical University of Vienna, University of Glasgow, UK, University of Wiscousin, Kyushu University, Monash University, Australia, Swinburne University of Technology, Australia, Kogakuin University, Faculty of Tropical Medicine, Mahidol University, Faculty of Medical Technology, Mahidol University, Faculty of Science, Mahidol University, Faculty of Medicine Siriraj Hospital, Mahidol University, Faculty of Medical Ramathibodi Hospital, Mahidol University etc.

Above are suggestions and past internship information, you can find out a list for internship with administrator team. A student who registers for Biomedical Engineering Training (EGBI 391) must obtain not less than 240 hours of training during the summer semester at the third-year of study. The training schedule will be five days a week in total of 240 hours or above. During the internship, you also need to contract your project supervisor for documenting. Also, they can write you a recommendation letter for you internship application. Lastly, Mahidol university also offer the scholarship for student mobility to support your international experience. you can subscribe for this opportunity through this Website or contract your administrator team.

3.4.3 Capstone Project or Research Work

During your study at fourth year, you need to register for capstone project course for final year project (FYP). Project or research work must follow the time as indicated in the course syllabus each project or research work will have 2-3 students. Each group will be advised by the faculty in the department or faculty that are selected from the curriculum committee. Project or research work of biomedical engineering is conducted to generate new knowledge and solve the biomedical engineering problem based on the integrated knowledge from different areas. Project or research work of biomedical engineering is carried out under supervision of an Biomedical Engineering faculty members which are designed as project advisor as well as examination committee.

The standard of leaning outcome of project or research work of biomedical engineering aims to gain various knowledge and skills in different aspects including 1) problem solving skills, 2) analysis and synthesis in the engineering design process, 3) develop and conduct appropriate experimentation, analyze and interpret data, and draw conclusions, 4) communicate effectively with a range of audiences, 5) recognize ethical and professional responsibilities, 6) demonstrate awareness in the ongoing need for additional knowledge, 7) function effectively on teams that establish goals, plan tasks, meet deadline and 8) function independently and self-directed to explore knowledge and technology. There will be designated faculty members for each students or advisor who will help mentoring each student. Advisors will schedule an appointment to follow up the progress. The evaluation processes for each semester in Plan A and B will be conducted by the department of Biomedical Engineering, Mahidol University. The following consideration will be used; i) Evaluate the quality and progress of the project; ii)Evaluate student work and outcome; iii) Evaluate the overall performance of student by oral examination; iv) Evaluate the final report

Every student must select a faculty advisor for the undergraduate engineering project. The project topic is announced beforehand and is based on its potential to enhance the student's knowledge. The evaluation processes for each semester in Plan C will be conducted by the department of Biomedical Engineering, University of Strathclyde

3.4.4 English Proficiency Requirement

Mahidol University has established the following English proficiency standard scores for undergraduate students at Mahidol University to ensure academic success. Students must meet one of the following criteria before end of the study:

- **MU-ELT:** Score of 84 or above

- **MU GRAD test:** Score of 70 or above
- **TOEIC:** Score of 600 or above
- **TOEFL ITP:** Score of 500 or above
(except for Mahidol University International College, or MUIC)
- **TOEFL iBT:** Score of 64 or above
- **MU-ELT:** Score of 42 or above
(for those who are graduates or prospective graduates of a bilingual/English program)
- **IELTS:** Score of 5.0 or above
- **Duolingo English Test:** Score of 90 or above

Additional Notes: For new students starting in the academic year 2023 (Thai curriculum), English scores are required for placement in the first-year English group. Students must submit their scores within 2 years from the date of enrollment. The English test is available via the student portal or website https://smartedu.mahidol.ac.th New students who do not have scores or do not meet the MU-ELT criteria must take the MU-ELT test and Mahidol University will allow the scores to be submitted. For more details, please contact the Educational Administration Division at Mahidol University at 02-849-4571 or visit https://smartedu.mahidol.ac.th

Additional Information: Students enrolled in the high standard program must provide the required English proficiency scores. If students meet the standard scores for both compulsory and elective English subjects, they are exempt from the English I and II courses. If you have submitted the score for exemption, please wait for confirmation from the university. For further information of English Profieciency Test, please visit website https://mustudent.mahidol.ac.th/2023/06/30821/.

3.4.5 Activity Transcript

AT or Activity Transcript is the certificate for extracurricular activities. AT system will record activity hours of students until they graduate. Students can check their AT progress at www.activity.mahidol.ac.th and they will get AT certificate to use as their employment application and etc.

In order to allow students to have an opportunity to learn and gain experience from outside of the classroom and foster them to become graduates who portray desirable characters and organizational culture of Mahidol University; and to ensure that extracurricular activities provided for Mahidol University's students are conducted in an orderly manner, Mahidol University requires the students who enrolled in the 2017 academic year onwards to participate in extracurricular activities provided by the University alongside their study according to the details as follows:

1. Students must participatein the activities for **at least 100 participation hours**. See the attachment for the structure of the activities. Students who fulfill the requirements will receive the Activity Transcript from the university upon their graduation.

2. Undergraduate students in regular programme are advised to be a member of at least one student club. The membership status will be included in their Activity Transcript.

3. Graduate students participate in extracurricular activities according to the Graduate School's requirement.

4. Regarding the students' enrollment prior to the 2017 academic year, the students must follow the Announcement on the requirements for students to participate in Mahidol University students'extracurricular activities previously issued prior to this current Announcement. In cases when the Announcement is not applicable, a decision is upon the discretion of the vice president in charge of student affairs, and the decision is considered final

3.5 Grading and Evaluation system

The grading system is according to the criteria from Mahidol University's regulations on undergraduate studies and the faculty of engineering's regulations and/or announcements.

1. Total credits and study duration For undergraduate program (4 years), the total number of credits are not less than 120 credits in semester system. The maximum duration is not more than 8 years started from the first year of admission.

2. Details of semester system - One academic year is divided into 2 semesters, Each semester comprises of at least 15 weeks. One credit of course (theory) comprises of 1) 1 hour per week or at least 15 hours in one semester of lecture, problem discussion time or equivalent and 2) 2 hours per week or at least 30 hours in one semester of self-study. One credit of course (Laboratory) comprises of 1) 2-3 hours per week or 30-45 hours in one semester of laboratory time or equivalent and 2) 1 hour per week or at least 15 hours in one semester of self-study.

3. Grade symbols with associated scores and no associated scores

Table 3.4: Grade symbols with scores and no association to the scores

Grade Symbols	Scores
A	4.0
B+	3.5
B	3.0
C+	2.5
C	2.0
D+	1.5
D	1.0
F	0.0

Grade Symbols	Description
AU	Audit
I	Incomplete
P	In Progress
S	Satisfactory
T	Transfer of Credit
U	Unsatisfactory
W	Withdrawal
X	No report

3.6 Student Portal

Mahidol has established a student portal for the student for the internal services. Here is the list of website that student should subscribe and familiar. Student also requires to keep up-to-date information.

- Student information and Student Services
  These services included, Fill out personal information Register for classes, Pay tuition fees, Check academic results, Book a dormitory, Register as a graduate, and etc. https://smartedu.mahidol.ac.th/Authen/login.aspx

- Student Information This section included: Academic calendar, Registration schedule, Announcements regarding education, New student activities, Student handbook http://www.student.mahidol.ac.th/

3.7 Student Username and University Internet

Students can obtain the Mahidol University network access code for logging into the Internet, MU Wi-Fi, the university email service, and educational systems. To activate an account, visit the Mahidol Internet account portal, select the "For Students" menu, and choose "Activate Account." Students should enter their identity verification information to receive the network access code; this is a one-time procedure applicable to all systems.

Student email usernames use the first name followed by the first three characters of the surname, for example `firstname.sur@student.mahidol.ac.th`. The address remains valid throughout the student's studies. After graduation, the domain changes to `@alumni.mahidol.ac.th`, for example `firstname.sur@alumni.mahidol.ac.th`. Student identifiers such as `u6600123` or `g6600123` may also be used when specified by the university.

Note that: E-mail Address is the service from Microsoft Office 365 for Education. That means you can download microsoft office software through IT system.

The Mahidol University network access code is personal information. Any actions performed using this access code are considered as actions done on behalf of the student registered at Mahidol University. Therefore, students should keep their password confidential, not disclose it to others, and change it regularly to prevent damage or violations under the Computer Crimes Act B.E. 2550 and the Computer Crimes Act (No. 2) B.E. 2560. Additionally, the access code is crucial throughout your studies at Mahidol University, as it serves as the key to accessing various university information systems, such as Internet communication networks (LAN and Wi-Fi), VPN from outside Mahidol University, and educational systems like e-Student, e-Evaluation, MUx, and We Mahidol Application

For IT-related issues, please send an email to consult@mahidol.ac.th. For problems with Username and Password, please email account@mahidol.ac.th.

3.8 Student's Dress Code

Mahidol student always requires to dress properly to attend the class. There are three dress code for Mahidol University

- **Formal attire**
  refers to the clothing that students wear on occasions involving ceremonies at Mahidol University, such as orientation for new students, teacher worship ceremonies, graduation ceremonies, state ceremonies, or as specified by Mahidol University

(a) Formal Dress

(b) Casual Dress

(c) Workshop Dress

Figure 3.1: Dress Code for Biomedical Engineering Student

- **Casual attire**
  refers to the clothing that students wear during classes, exams, or when interacting with university departments for work-related matters at Mahidol University.
- **Workshop attire** refers to the clothing that students wear during workshop class, not examination, or enter the lab area, or when interacting with university departments for work-related matter in the workshop area.

This dress code requires from the university and faculty. The main purpose of this dress code is to improve your professions and ensure of your safety. The staff or lecturer may be ask you to leave, if you not wear it properly. Please refer dress code to website Student Dress Code Links

Note! The student can wear the student uniform comply with your sexual identity, sexual Physical, or Indirect sexual physical.

Chapter 4

Research Tracks

Welcome to our research labs! As an undergraduate student, you'll have the exciting opportunity to explore various advanced fields in biomedical science and technology. Our department is home to several specialized laboratories where cutting-edge research takes place. There are 6 research tracks in our department:

1. **Neuroengineering**
   Track Leader: Soontorn Oraintara, Ph.D.
   Email: soontorn.ora@mahidol.ac.th
   Description: Understand the nature and data acquisition of biomedical signals such as EEG, EMG, and ECG as well as medical images such as MRI, fMRI, and CT. Investigate and study signal and imageprocessing theories related to biological and medical applications. Apply signal and image processing knowledge to enhance clinical analysis and develop the solid background on implementing clinical instruments.

2. **Drug delivery system and tissue engineering**
   Track Leader: Norased Nasongkla, Ph.D.
   Email: norased.nas@mahidol.ac.th
   Description: This research track consists of 2 major topics; Tissue Engineering and Drug Delivery System. The Tissue Engineering research involves with the scaffold preparation for tissue engineering application. We are currently focusing on the modification of Silk fiber to be use as biomaterials. The examples of areas we studied are; (1) Surface chemistry of the biomaterials, (2) Interfacial study of Biological Tissue and (3) Adhesion mechanism of the various cells on various subtracts. The Drug Delivery System research has its objective to develop polymeric drug delivery systems (DDSs) that can efficiently deliver anticancer drug to cancer cells. Polymers used for these DDSs are biocompatible and biodegradable. DDSs can be prepared in millimeter, nanometer and molecular scale. The applications are, such as, development of intratumoral implantations, development of magnetic nanoparticles for MRI detection and various sizes of anticancer drugs.

3. **Advanced Computing in Medicine**
   Track Leader: Panrasee Rittapravat, Ph.D.
   Email: panrasee.rit@mahidol.ac.th
   Description: This track aims to conduct research related to developing and applying advanced computational technologies to medical applications, such as mathematical model development in physiology, medical image processing and visualization, medical cloud computing management, telemedicine etc. Examples of research projects in this track include development of medical image segmentation and 3D visualization techniques for nasopharyngeal carcinoma, telemedicine for brain cancer diagnosis.

4. **Rehabilitation Engineering and Artificial Organs**
   Track Leader Warakorn Charoensuk, Ph.D.

Email: warakorn.cha@mahidol.ac.th
Description: Rehabilitation Engineering and Artificial Organs are a growing specialty area of biomedical engineering. Rehabilitation engineers enhance the capabilities and improve the quality of life for individuals with physical and cognitive impairments. They are involved in prosthetics, the design of assistive technology that enhances seating and positioning mobility, and communication. Artificial Organs applies multi-knowledge of both engineering and medical science to improve the quality of life and treatment. It includes the study of materials, biological function, and transport of chemical constituents across biological and synthetic media and membranes materials. Progress in biomechanics has led to the development of the artificial heart and heart valves, artificial joint replacements, as well as a better understanding of the function of the heart and lung, etc.

5. **Medical Robotics**
Track Leader: Jackrit Suthakorn, Ph.D.
Email: jackrit.sut@mahidol.ac.th
Description: This track can be categorized into two fields; Medical Robotics and Computer-Integrated Surgery. There is a common area between both fields. Computer-Integrated Surgery (CIS) is known in various names, such as, Computer-Aided Surgery, Computer-Assisted Surgery. CIS is a multi-disciplinary which based on current cutting edge technologies which are available in medical field, especially, advanced imaging systems. CIS is split into 3 sub-procedures; 1) Pre-Operation, 2) Intra-Operation and 3) Post-Operation. The pre-operation tasks in CIS involve, e.g., surgical training, medical image enhancement, surgical planning, procedure planning and tool selecting based on computerize systems or processes. Therefore, this process is similar to CAD – Computer-Aided Design in engineering process. The intra-operation tasks in CIS involve surgical executions, real-time surgical tool tracking, tool and image registrations, surgical navigations, robot-assisted surgery, image-overlay techniques and more. However, this stage is the most critical one since the systems should be flexible, adjustable and repeatable. This process is similar to CAM – Computer-Aided Manufacturing in engineering process. The post-operation tasks would relate to surgeon's performance checking which is similar to TQM – Total Quality Management in engineering process. Medical robotics can be a part of CIS while various medical related applications also available.

6. **Biosensors and Medical Instrumentation**
Track Leader: Chamras Promptmas, Ph.D.
Email: chamras.pro@mahidol.ac.th
Description: Biosensor is an analytical device which converts physiochemical changing response into electrical signal. The changing response is a specific reaction between substrate and biorecepter. The biosensor is often used to determine the parameter or concentration of interest biological substance. These sensors mostly used in many applications such as medical, industrial, military and environment.

4.1 List of Research Laboratory

Biomedical engineering provides you various research field for the cutting edge technology for biomedical applications. Below are research labs associated with our department

1. **Center for Biomedical and Robotics Technology Laboratory [BARTLAB]**
   Director: Assoc. Prof. Jackrit Suthakorn, Ph.D.
   Other Academic Staff: Shen Treratanakulchai, Yuttana Itsarachaiyot, Nantida Nillahoot, Choladawan Moonjaita
   Contact: info@bartlab.org, jackrit.sut@mahidol.ac.th
   Address: Building 2, Faculty of Engineering
   Website: www.bartlab.org
   Short Description: Center for Biomedical and Robotics Technology (BART LAB) is a collaborative medical/engineering research center for medical robotics and computer-integrated surgical technology with missions to conduct research, to develop and commercialize prototypes, and to create research and knowledge networks in the areas. The goal of BART LAB is to increase the ability, potential and efficiency of the research and development on various fields in Biomedical Engineering Technology in Thailand. This also responds to the government's R&D strategies.

2. **Brain-Computer Interface Laboratory [BCI]**
   Director: Soontorn Oraintara, Ph.D.
   Other Academic Staff: Jetsada, Titipat
   Contact: soontorn.ora@mahidol.ac.th
   Address: Building 2, Faculty of Engineering
   Website: -
   Short Description: Description: Brain computer interface lab conducts research in general areas of brain computer interface (BCI), biomedical signal/image processing, genomics signal processing, bioinformatics, pattern recognition, time-frequency analysis, and data compression.

3. **Smart Motion Analysis and Rehabilitation Technology [SMART LAB]**
   Director: Asst. Prof. Warakorn Charoensuk
   Other Academic Staff: -
   Contact: warakorn.cha@mahidol.ac.th
   Address: Building 2, Faculty of Engineering
   Website: -
   Short Description: SMART LAB aims to analyze human motion and develop devices and computer program for analyzing and improving the quality of life for those who have movement problems.

4. **Artficial Intelligence in Medicine [AIM LAB] Director**: Assoc. Prof. Panrasee Ritthipravat, Ph.D.
   Other Academic Staff: -
   Contact: panrasee.rit@mahidol.ac.th
   Address: Building 3, Room , Faculty of Engineering
   Website: https://www.eg.mahidol.ac.th/dept/egbe/webaim/index.html
   Short Description: Our research objective is to develop and apply artificial intelligent techniques to various medical applications. ongoing research includes developing software for prediction of cancer recurrence, image segmentation, 3D visualization of tumor extent and robot-assisted developmental stimulation for autistic children etc. AI techniques employed in these research projects include neural networks, fuzzy logic, probabilistic reasoning, genetic algorithm, and hybrid approaches etc.

5. **Biopolymers and Nanoengineering for Drug Delivery and Molecular Imaging Lab [BioNEDD LAB]**
   Director: Prof. Norased Nasongkla, Ph.D.
   Other Academic Staff: Chalaisorn
   Contact: norased.nas@mahidol.ac.th
   Address: Building 3, Room , Faculty of Engineering
   Website: http://www.bionedd.org/
   Short Description: Our research objective is to develop polymeric drug delivery systems (DDSs) that can efficiently deliver anticancer drug to cancer cells. Polymers used for these DDSs are biocompatible and biodegradable. DDSs can be prepared in millimeter, nanometer and molecular scale. For the millimeter scale, the research aim is to develop intratumoral implantations that can be injected directly insides tumors. Anticancer drugs that are kept inside will release and kill cancer cells. For the nanoscale, we develop magnetic nanoparticles that can be detected by MRI once injected inside the body. Not only to visualize cancer cells in vivo but these nanoparticles can also selectively deliver anticancer drugs to these cancer cells. For the molecular scale, anticancer drugs are chemically conjugated to each polymer chain to enhance the cancer cell uptake and reduce the side effects of these drugs.

6. **Micro Total Analysis System [MicroTAS LAB]**
   Director: Asst. Prof. Chamras Promptmas, Ph.D.
   Other Academic Staff: -
   Contact: chamras.pro@mahidol.ac.th
   Address: Building 3, Room , Faculty of Engineering
   Website: -
   Short Description: This research track consists of 2 major integrated technologies: Microfluidics Technology and Detection Technology. Using existing materials and fabrication methods establish microfluidic channel network connecting functionalized compartments to improve the ease-of-use of μTAS. Detection technology can be chemical or biosensors embed in the system. This track applies several detector such as optical device, piezoelectric crystal, micro-cantilever, ion-sensitive field effect transistor (ISFET) and others.

7. **Biomedical Cloud Computing LAB**
   Director: Songpol Ongwattanakul, Ph.D.
   Other Academic Staff: -
   Contact: songpol.ong@mahidol.ac.th
   Address: BART LAB
   Website: -
   Short Description: We introduce an application of cloud computing technology for big data processing from biosensors that integrates with multidisciplinary experience to resolve complex medical problems.

8. **Machine Intelligence & Neural Technology Laboratory [MINT LAB]**
   Director: Jetsada Arnin, Ph.D.
   Other Academic Staff: -
   Contact: jetsada.arn@mahidol.ac.th
   Address: Building 2, Faculty of Engineering
   Website: -

Short Description: Machine Intelligence & Neural Technology Laboratory (MINT LAB) MINT Lab fuses biomedical engineering with machine intelligence, focusing on signal processing, interconnected IoT development, and data-driven healthcare innovations. We're reshaping the future of medical technology.

9. **Biosensors Laboratory**
   Director: Asst. Prof. Benchaporn Lertanantawong, Ph.D.
   Other Academic Staff: -
   Contact: benchaporn.ler@mahidol.ac.th
   Address: Building 3, Faculty of Engineering
   Website: -
   Short Description: Our group researches on the electrochemical, chemical, and physical platforms combined with the use of biomolecules such as DNA/RNA, antibody, enzymes to develop biosensors. We aim to create the POCT, mostly DNA sensors to achieve the highly sensitive and simultaneous detection for medical application.

10. **Ultrasonic Characterization of Materials Properties [UltraCMP]**
    Director: Pracha Yambangyang, Ph.D.
    Other Academic Staff: -
    Contact: pracha.yam@mahidol.ac.th
    Address: Building 3, Faculty of Engineering
    Website: -
    Short Description: UltraCMP lab aims to detect and characterise damage or defects in materials by using linear and nonlinear ultrasonic techniques. The application relates to detect small scale defects or structural degradation, such as micro cracks, and material characterisation in the deep tissue injury.

11. **Advanced Molecular Diagnostics [AMD LAB]**
    Director: Pornpat Athamanolap, Ph.D.
    Other Academic Staff: -
    Contact: pornpat.ath@mahidol.ac.th
    Address: Building 3, Faculty of Engineering
    Website: -
    Short Description: AMD lab conducts interdisciplinary research by integrating molecular biology, engineering, and also computer science (e.g., artificial intelligence) for various medical applications. Our aim is to develop novel diagnostic tools for improving early detection and treatment of diseases, such as infectious diseases and cancer.

12. **Biomedical and Data Lab [BioDat LAB]**
    Director: Titipat Achakulvisut, Ph.D.
    Other Academic Staff: -
    Contact: titipat.ach@mahidol.ac.th
    Address: Building 2, Faculty of Engineering
    Website: -
    Short Description: Our lab work in an intersection of applied natural language processing, machine learning, and science of science. We aim to build tools to make better science. We also broadly interested in machine learning and deep learning applications for bioengineering and biomedical science.

List of Lecturer in department of Biomedical Engineering is shown in table 4.1.

Table 4.1: List of Biomedical Engineering Lecturer

Name	Research Interests	Lab Name
Jackrit Suthakorn, Ph.D. jackrit.sut@mahidol.ac.th	Leader in Medical Robotics: Robot-Assisted Surgery, Surgical Navigations	BART LAB
Choladawan Moonjaita, M.Eng. Choladawan.moo@mahidol.ac.th	Medical Robotics, Biomechanics	BART LAB
Yuttana Itsarachaiyot, Ph.D. yuttana.Its@mahidol.ac.th	Computational Robotics, Medical Robotics, Continuum Mechanics	BART LAB
Shen Treratanakulchai, Ph.D. shen.tre@mahidol.ac.th	Medical Robotics, Soft Robotics, Manipulator, Mechatronics	BART LAB
Songpol Ongwattanakul, Ph.D. songpol.ong@mahidol.ac.th	Biomedical Cloud Computing	Biomedical Cloud Computing LAB
Soontorn Oraintara, Ph.D. soontorn.ora@mahidol.ac.th	Biomedical signal and image processing, Modeling of biomedical signals and images	BCI LAB
Warakorn Charoensuk, Ph.D. Warakorn.cha@mahidol.ac.th	Human Motion Analysis and Rehabilitation Technology, Mixed reality- (MR) based health-related physical fitness assessment system, Prostheses and Orthoses	SMART LAB
Panya Kaimuk, M.D. Panya.kai@mahidol.ac.th	Sports Science, Golf, Orthopedic Surgery	SMART LAB
Norased Nasongkla, Ph.D. Norased.nas@mahidol.ac.th	Leader in Tissue Engineering and Drug Delivery Systems Research Track	BioNEDD LAB
Chalaisorn Thanapongpibul, Ph.D. chalaisorn.tha@mahidol.ac.th	Nanotherapeutics, Smart nanoparticles, Advanced technologies in nanoparticle synthesis	BioNEDD LAB
Pinunta Nittayacharn, Ph.D. pinunta.nit@mahidol.ac.th	Ultrasound contrast agents, Image-guided drug delivery, Cancer detection and therapy, Nanomedicine	BioNEDD LAB
Chamras Promptmas, Ph.D. chamras.pro@mahidol.ac.th	Director in Biosensors and Micro Total Analysis System Laboratory	MicroTAS LAB
Pracha Yambangyang, Ph.D. pracha.yam@mahidol.ac.th	Director in, Biomedical instrumentation Laboratory, Biomedical Instrumentation, Clinical Engineering, Hospital Engineering	UltraCMP
Benchaporn Lertanantawong, Ph.D. Benchaporn.ler@mahidol.ac.th	Nanobiosensors for biomedical application, DNA self-assembly, Electrochemistry, Nanomaterials for sensing application	Biosensors LAB
Jetsada Arnin, Ph.D. jetsada.arn@mahidol.ac.th	Neuro-Rehabilitation, Wireless Embedded Systems	MINT LAB
Pornpat Athamanolap, Ph.D. pornpat.ath@mahidol.ac.th	Molecular Diagnostics, Microfluidics, Point-of-Care Diagnostics, Infectious Diseases, Epigenetics	AMD LAB
Titipat Achakulvisut, Ph.D. titipat.ach@mahidol.ac.th	Applied Machine Learning, Natural Language Processing, Metascience	BioDat LAB
Theeraporn Rubcumintara, Ph.D. theeraporn.rub@mahidol.ac.th	Material Engineering, Biomaterial	

4.2 Student Research Experiences and Extra Curriculum

student support services are shown to be available to improve learning experience and employability.

The programme encourage and support student to have Co-curricular activities, student competition, and other student support services to improve their learning outcome and employability.

1. Co-curricular activities play a major role in development and enhance academic experiences for undergraduate students. some co-curricular activities is listed below

    (a) **Research and publication:** Our undergraduate Study course encourage student to publish in international peer review paper. The students that get involved with international conferences or journal help them engage in research activities related to field of study. our undergraduate student has a chance to conduct their research from their original idea or select on-going project working closely with their supervisor. Most of our research is funded by national

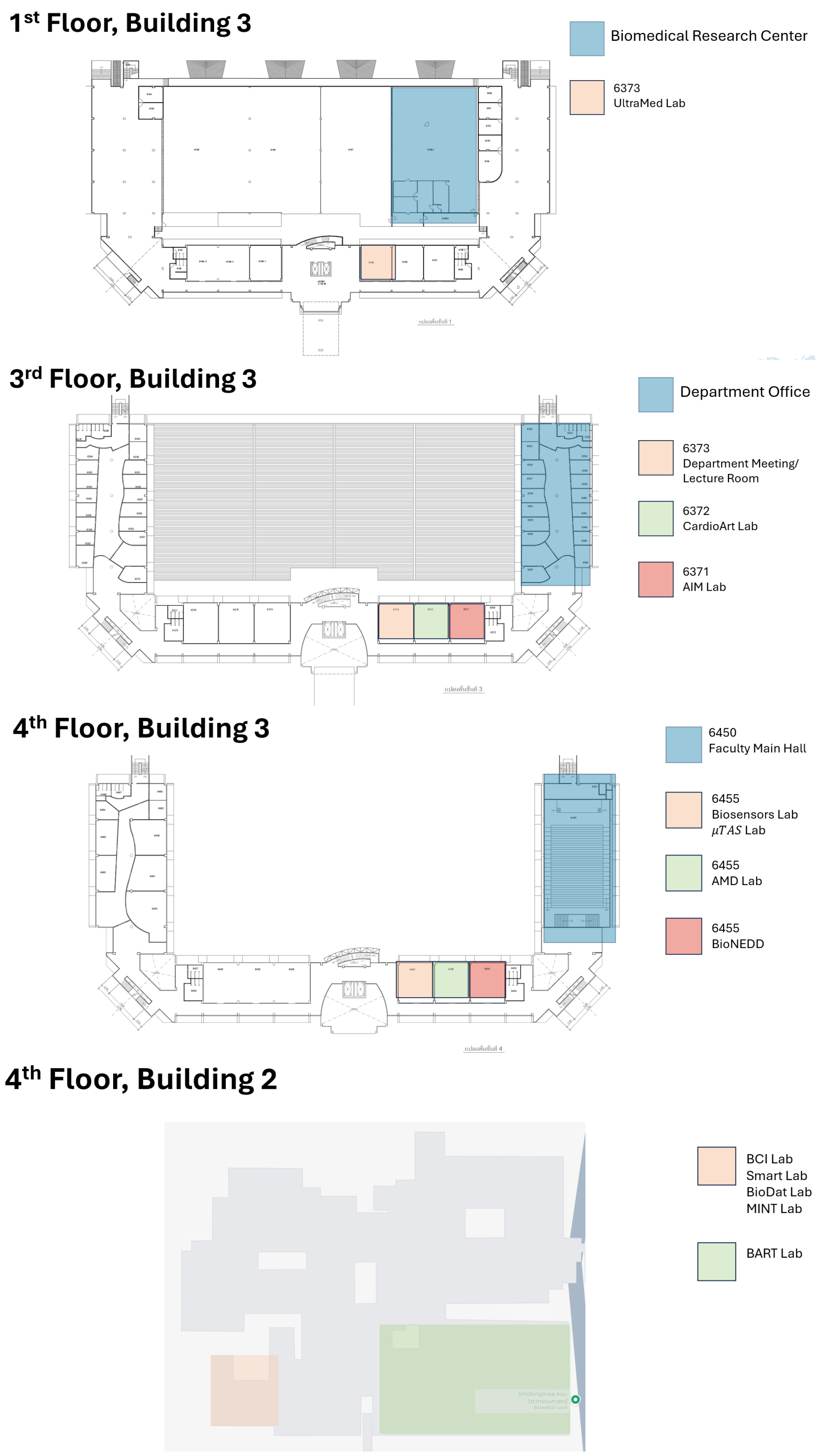

Figure 4.1: Maps and Lab Space

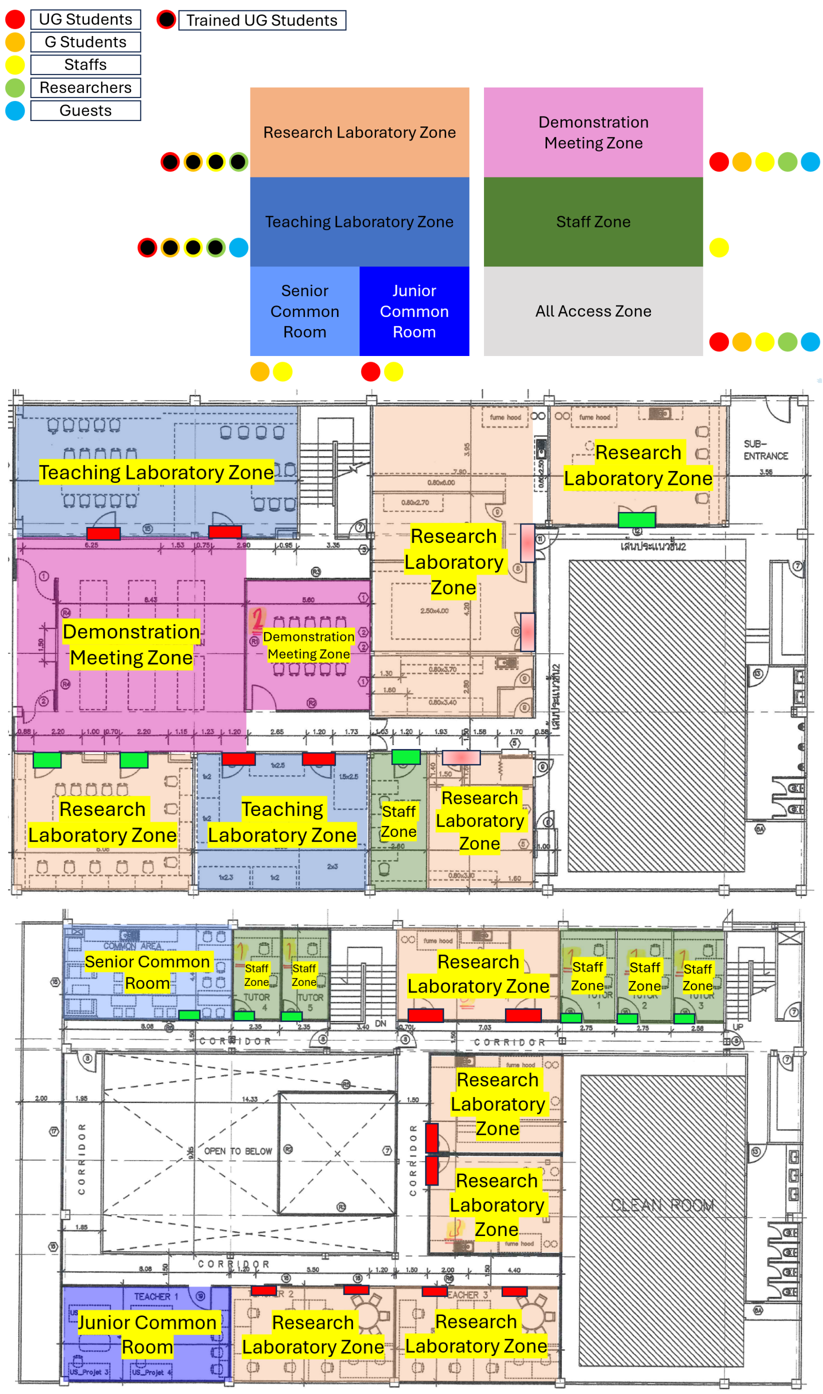

Figure 4.2: Biomedical Research Center access zones

or government agencies which could be foundation research or applied science research. We expect that the student to present their findings at conference and publish in academic journal at least. We also teach them how to write project proposal for their future career in academic.

(b) **Professional development and workshop:** university and department provide a wide range of workshop and seminar focus on professional skills that are relevant to the undergraduated students' future career. This workshops cover areas such as communication skills, presentation skills, grant writing, project management, entrepreneurship, and networking. Department set the schedule for seminar in biomedical field in every week and also invite guest speaker from outside institution or international guest to visit the laboratory or comment on student works.

(c) **Student clubs and associations:** there are number of clubs in faculty and university level. The undergraduated student can participate those clubs. These are the platform for networking, collaboration, and socializing among student. The member of each club can organize events, seminars, and social activities tailored to the interest and needs of student. For the club in faculty level, the student can see the list of the club via this link. For the club in university level, there are many clubs that cover areas in volunteering, art and cultural, sport, academic, and etc. please use this link to refer to the list of clubs.

(d) **Leadership development programs:** Underdgduate students can benefit from participating in leadership development programs or workshops. These programs can help them develop their leadership skills, project management abilities, and teamwork capabilities. Leadership development programs can help undergraduate students develop the skills they need to be effective leaders. These skills include communication, decision-making, problem-solving, and conflict resolution. Project management workshops can help undergraduate students develop the skills they need to successfully manage projects. These skills include planning, organizing, and executing projects. Teamwork workshops can help undergraduate students develop the skills they need to work effectively with others. These skills include communication, collaboration, and conflict resolution.

2. Student competitions: Our programme encourage the student to participate in wide range of competition. There are many forms of competitions such as research based competition, innovation competition, idea pitching, business plan competition, hackathorn, engineering and design competitions. These opportunities can provide valuable experience for academic and professional growth.

    (a) **Business plan competitions** are a great way for undergraduate students with entrepreneurial aspirations to develop their business skills and get their ideas off the ground. These competitions challenge students to develop innovative business ideas, create comprehensive business plans, and pitch their ideas to a panel of judges. Winners often receive funding, mentorship, or networking opportunities to help turn their ideas into reality.

    (b) **Hackathons** are events where undergraduate students can showcase their programming skills and problem-solving abilities by developing software solutions or prototypes within a limited timeframe. Participants can compete individually or in teams, and judges assess the quality and functionality of their creations.Hackathons can be a great way for undergraduate students to develop their programming skills, problem-solving abilities, and teamwork skills. They can also be a great way to network with other students and professionals in the technology industry.

(c) **Engineering or design competitions** are a great way for undergraduate students to showcase their engineering and design skills. These competitions challenge students to design and build innovative solutions to specific problems. The solutions can be prototypes, sustainable designs, or solutions to engineering challenges. Judges evaluate the functionality, creativity, and practicality of the students' designs.

(d) **Innovation and technology competitions:** are a great way for undergraduate students to showcase their ideas, prototypes, or projects related to emerging technologies. These competitions often focus on fields such as artificial intelligence, robotics, biotechnology, or renewable energy. Judges assess the innovation, market potential, and practicality of the students' concepts.

(e) **Social entrepreneurship competitions:** Encourage undergraduate students interested in social entrepreneurship to participate in competitions that focus on creating innovative solutions to social and environmental challenges. These competitions recognize ventures with a social impact and provide support, mentorship, and funding opportunities to help students launch their initiatives.

(f) **Literary or creative arts competitions:** Promote literary or creative arts competitions where undergraduate students can submit their written work, poetry, visual arts, or performances. These competitions showcase students' creativity and talent while allowing them to connect with a wider artistic community.

The supervisor will help and provide adequate support, mentorship, and resources to help undergraduate students prepare for these competitions. Additionally, offer feedback and opportunities for improvement to encourage continuous growth and development.

Chapter 5

Student Support

5.1 Key Contacts and Resources

the University and department provide various forms of student support to ensure the well-being and success of students. the following list is the essential types of student support that university and department providing:

1. Academic Support: we have academics support to excel the student.

    (a) **Tutoring program:** the department has assign personal tutor and supervisor to help the student develop academically, personality and professionally throughout study. Personal tutor can offer you support and signpost you to specific service when student are experiencing any difficulties. Supervisor is an academic member in student department who support you during student studies and provide feedback of student work. Senior tutor is assigned by the department. These person has overall responsibility for pastoral care of student and give additional guidance and oversight in more complex problems. Faculty senior tutors are work closely to university and ensure delivery of high quality and consistent support for student.

    (b) **Language and Centre for English for communication** The Center for Translation and Language Services was established as part of the Master of Arts program in Language and Culture for Communication and Development. It has been providing translations for more than 10 years in the realization that we now live in a global world in which it is necessary to communicate in many other languages for academic, business, and social purposes, etc.

        The Center for Translation and Language Services is a cross- culture communication center that aims to facilitate modernization through communication and exchange of information and understanding at an international level. Our services aim to produce academic translations, including translating training courses, research translations, connecting with translator networks and expanding academic translation coordination with other Thai and foreign institutes. Besides that, the center also translates academic documents into other languages and offer proof-reading and editing by language experts in Thai, Bahasa Malay, Burmese, Vietnamese, Khmer, Lao, Chinese, Japanese, Korean, Arabic, English, French, German, Spanish, Portuguese, Italian and Russian. The full detail for the language service can be found in this https://lc.mahidol.ac.th/en/

    (c) **Student Representatives:** each cohort will have their elects student representatives who coordinate with staff closely to shape and improve student experience. the student can use this channel to feedback about department course, other aspect of life in department. Their representatives have opportunities to provide student feedback anonymously where appropriate.

(d) **The department and faculty skill workshop:** the university provides workshop to improve student skill that can be useful for their study for example computer programming, statistics, entrepreneuship, and management.

(e) **Library services:** this service offer workshop and guidance to support student academic life. the area of service include subject librarian guidance, searching for high-quality information via specialist, reference management, plagiarism awareness and copyright guidance. They also offer support for researcher in open-access and research data management. Student also can download proceedings or journal publication through Mahidol e-Database. The link of the MU Library can be found at this link https://www.li.mahidol.ac.th/

(f) **Life long learning:** This is a service offer online course and Literacy called MUx program. This program of various online contents include digital literacy, communication literacy, finance and management, 94 MOOC courses, and 421 Courses.

2. **Career services:** This service provide guidance and assistance to prepare their future careers. This service offer counselling, resume writing workshop, Jobs interview skill, internship and job placement assistant, career fair, and networking event. This service ensure that student is well prepared for their future career in both industrial and academia. Career services offer counseling, resume writing workshops, mock interviews, internship and job placement assistance, career fairs, and networking events. Mahidol offer wide range career service. Full detail of career service can be found in https://careers.mahidol.ac.th/.

   (a) **career introduction**, this service will give you the information on each field of job including, politics, management, academic, tourism, business, journalism, health science, veteriranian, IT, science, and engineering. Career introduction by Alumni: This service provide you an information about work from specific career. An interviewing video will guide you about the job description and how to prepare student themselve for job interview.

   (b) **finding yourself service:** is the space for the student to explore themself for knowing your self

   (c) **Job searching and job fair:** This will gather the jobs advertisement and internship program for student. The university or faculty provide job fairs to invited the company and industry to university for active recruit our student for their work.

   (d) **MU Career advancement program MUCAP:** This program promote the relationship between current student and alumni through mentorin system. A mentor which is our alumni can give a guidance and recommendation to mentee which can improve skill from the realworld. There are sub-project under this program including MUCAP mentee interview, Batch orientation, mentoring session, career round table, and MU farewell.

3. Wellbeing service: During studies in Mahidol, there might a range of practical issue in living and study that student need to take care of, for example, managing their money, finding accommodation, VISA, or even management family. The university provide information and advice of a range of common issues and direct you where you can access expert support and advice.

(a) **VISA:** this service ensure that student have a correct VISA for the duration of their studies and be a point of contract to the immigration officer. They will give you useful information and evidence support to the immigration officer to grant the student VISA. The student is required to report their stay every 90 days in Thailand, the VISA team will notify you about VISA reporting. Not only the VISA for the study, this also including the VISA for the graduation ceremony.

(b) **Accommodation:** National and international student might have a difficulties in renting the accommodation during their student. The advisor can support them in finding the right house solution. The university also offer a list of apartment complexes on campus. The condominium at Mahidol Salaya (Condo A for male and Condo D for female). The room has bathroom, AC, water supply, and WIFI. the rent is start from 5,500 Thai Baht PCM(Per calendar month) or 800 Thai Baht. This rent is not include the electricity and water. Around Salaya campus which is a base of department has a numerous of new accommodation around the campus. The student can check the list in https://mahidol.ac.th/accommodation/.

(c) **Money:** When the student studying full-time or part-time, it can be expensive. There are number of solutions that student can save the money as a student for example student discount, get a student travel card. shop for cheaper food brands. The university also has support for students who are experiencing financial hardship. The student can find more detail in https://op.mahidol.ac.th/sa/en/2021/12834/.

4. Disability support services: Disability Support Services unit, the Division of Student Affairs is set up for helping and supporting disability students to equivalent access university educational systems same as others, reducing barriers and providing personal essential support services to help disability students achieve their educational goals. Also, create community that disability students can live along with other students well. There are various types of support; 1) Facilitate the exam such as extended time or read the exam paper; Produce learning media such as braille documents, media images, and enlarged documents; 3)Facility Technology Services; 4) Volunteer services; 5)Counseling services; and 6) Translation and narrator services. Two locations for student service disability are located at a) Disability Support Services Unit, Division of Student Affairs, Mahidol University Service disability students from every college and faculty on any campus Location: Volunteer Center, 1st floor, Mahidol Learning Center, Mahidol university Salaya campus, and Ratchasuda College. For full detail please go to https://www.facebook.com/dssmahidol or email contract to mahidol.dss@mahidol.ac.th.

5. Mental health service: student with good mental health means being able to feel, think, and react in the way that allow student to live their live in the way they want to. At poor mental health period, student might find that student are frequently thinking, feeling or reacting become difficulties, or impossible, to cope with. If you experience symthoms such as depression, self-harm, anxiety, and eating disorders which are mental ill health. If student have already been diagnosed with a mental health problems. There are a number of service at Mahidol where you can access privately with non-judgemental help and support. As a way to relieve anxiety,

encourage self and others understanding, and solve problems with the right decision. Students are encouraged to become Mahidol University desired characteristics of the graduates and wisdom of the land. The service channels are as follows. Walk in as group or personal : Mahidol Friends, 3rd floor Mahidol Learning Center (to make an appointment please call 02-849-4538, 4502 or 4512). Telephone counseling : please call 02-849-4502 or 4512. MU Hotline (in case of emergency) : please call 088-874 7385 (24 hours). Inbox Facebook Fanpage : @MahidolFriends

6. Health services: Student can access to all kind health services such as stuent health care unit, sport clinic, adolescent clinic, tropical disease hospital, and etc., through our hospital and medical school. Mahidol University are cooperation with social welfare. Therefore, Mahidol students must show student card with identification card (or passport for foreign students) when they use health care services. Note: Some of medical expenses are exclude from the list established by Comptroller General's Department payment. Therefore, the students are responsible for those expenses.

7. Bullying and harassment: Mahidol University has a zero-tolerance policy towards bullying or harassment. If student encounter such behavior, there are various avenues available to you, providing advice and support to help you make informed decisions about your next steps. the student can report and get support through report and support system anonymously.

8. Technology Support: As technology plays a significant role in education, universities should provide technical support and resources for students. This includes computer labs, access to software and online learning platforms, IT assistance, and reliable Wi-Fi throughout campus. The student can make their request for the help regarding IT by https://muit.mahidol.ac.th/ website.

5.2 Student's Activity

Student activities are an integral part of the university experience, offering a wide range of opportunities for personal growth, skill development, and community engagement. These activities encompass various clubs, organizations, and events that cater to diverse interests and passions, from academic and professional societies to cultural, recreational, and social groups.

Academic and professional organizations provide platforms for students to deepen their knowledge in specific fields, network with professionals, and gain practical experience through workshops, seminars, and competitions. For instance, engineering students might join the robotics club, while business students could participate in finance and investment groups. These activities not only enhance learning but also help build resumes and prepare for future careers.

Cultural and diversity clubs celebrate the rich tapestry of backgrounds and traditions within the student body, promoting inclusivity and understanding. These groups often host events such as cultural festivals, heritage month celebrations, and international food fairs, fostering a sense of community and global awareness.

Recreational activities and sports teams offer a balance to academic life, promoting physical health and well-being. Whether through intramural sports, fitness classes, or outdoor adventure clubs, students can stay active, relieve stress, and build lasting friendships.

Social and volunteer organizations encourage students to engage with their communities and make a positive impact. From organizing charity drives and community service projects to participating in environmental sustainability initiatives, these activities cultivate a sense of responsibility and altruism.

Participating in student activities also develops essential life skills such as leadership, teamwork, time management, and communication. Leadership roles in student government, organizing events, or leading club projects provide practical experience that is valuable in any career path.

Overall, student activities enrich the university experience by offering opportunities for exploration, connection, and growth beyond the classroom. They help students build a well-rounded skill set, create lifelong memories, and prepare for a successful future.

5.2.1 Biomedical Engineering Yearly Activity

Our department has our own student activity. here is the list of the activity

- **Student orientation:** This is the welcome week which takes place the week before start of first semester
- **Internship Presentation:** our 4th year student present their internship experience from the last summer. This is an opportunity for bme student get to know what they will experience from their senior student at the first two weeks of September.
- **Graduation Ceremony:** Every year our department will celebrate our graduated BME student at first week of October.
- **Advisor and research selection:** around november each year, our third year student will go through the process of advisor selection for their capstone project in the next year.
- **BME Relationship:** BME student will participate an annual BME relationship sport event. This event the student will have an opportunity to get to know other BME student from other institution at the second week of January.
- **BME Playground:** this is a biomedical engineering camp for high school student to get to know our department. All BME student will set up the 3 day camp at the faculty to demonstrate and teach highschool student about our study here at BME Mahidol University at the last week of January or first week of February.
- **BME Sport day:** This is an internal sport day that all the BME member will come together and doing sport and celebrate a new year party on every February.

5.3 Travel to Mahidol Salaya

Mahidol University offers service for student who want to travel across the campus and travle inside the campus. There are 3 choices of service run by mahidol. Full detail can be founded in https://op.mahidol.ac.th/ir/th/life-at-mahidol-university/.

- **Shuttle Bus Service**
  Mahidol University provides a shuttle bus service for staff and students between its Salaya, Phayathai and Bangkok Noi Campuses. The service is free of charge.

- **Salaya Campus Tram Service**
  Salaya Campus is serviced by three tram routes (Route 1 – Green, Route 2 – Blue, and Route 3 – Red). These allow staff, students and visitors to commute around the campus. Trams are offered free of charge and are available from 6.30 A.M. – 8.00 P.M. on weekdays. A reduced service is provided at the weekend and public holidays.

- **Salaya Link**
  This bus service links Mahidol University (at the College of Music) to the Bang Wa Sky train (BTS) station. A one-way trip is 30 Thai baht (about US$1).

  For more information, please visit: http://www.music.mahidol.ac.th/salayalink.

for the public transportation and other mode of travel.

- **Buses** are a convenient way to travel between Bangkok and Salaya Campus. The air-conditioned buses No. 515 and No. 547 run on different routes from Phuttamonthon 4 Road to Victory Monument in central Bangkok. Regular bus No. 124 runs from Phuttamonthon 4 Road to Pinklao west of the Chao Phraya River (and continues into central Bangkok), connecting to the Express River Ferry at Pinklao Bridge Other Regular buses No. 125 and No. 84 also runs on different routes from Phuttamonthon 4 Road to Samsen Train Station and Klongsarn Pier. [... further information at Bangkok Mass Transit Authority (BMTA) ...]

- **Taxis** at Salaya are usually available at the small taxi stand on Phuttamonthon 4 Road in front of the campus. From Salaya expect to pay about 125 baht to Central Pinklao, 190 baht to Siam Square, 250 baht to Don Muang Airport and 800 baht to Suvarnabhumi Airport. Motorcycle taxis also wait here for short excursions.

- **Driving to Salaya Campus**– If driving from east of the Chao Phraya River (central Bangkok): cross the river westbound over either the Phra Pinklao Bridge or Rama 8 Bridge. These two routes meet in Pinklao; follow Phra Pinklao Road westward. Alternatively, cross Sang Hee (Krung Thon) Bridge from central Bangkok on Rajvithi Road, which becomes Sirinthorn Road and then meets the Phra Pinklao Road route in Taling Chan. All routes then follow Boromrajchonnee Road westward toward Salaya.

  Continue straight on Boromrajchonnee Road, following the signs to Nakhon Pathom or Nakhon Chaisi, and eventually the sign to Salaya. The exit to Salaya is on the left; once off Boromrajchonnee Road, stay in the right lane, which veers off to the right on a flyover, crossing Boromrajchonnee Road onto Phuttamonthon 4 Road. The well-marked gates of Mahidol University are immediately on the left after descending from the flyover.

  A toll-free elevated expressway with faster traffic also runs from the Pinklao area over Phra Pinklao Road and Boromrajchonnee Road most of the route to Salaya. The route coming from Rama 8 Bridge goes directly onto the elevated expressway while the other two routes begin at ground level.